

MySQL Project

E-COMMERCE CUSTOMER CHURN ANALYSIS

Submitted by: SHAHEEKA PK

Tool used: MySQL

INTRODUCTION

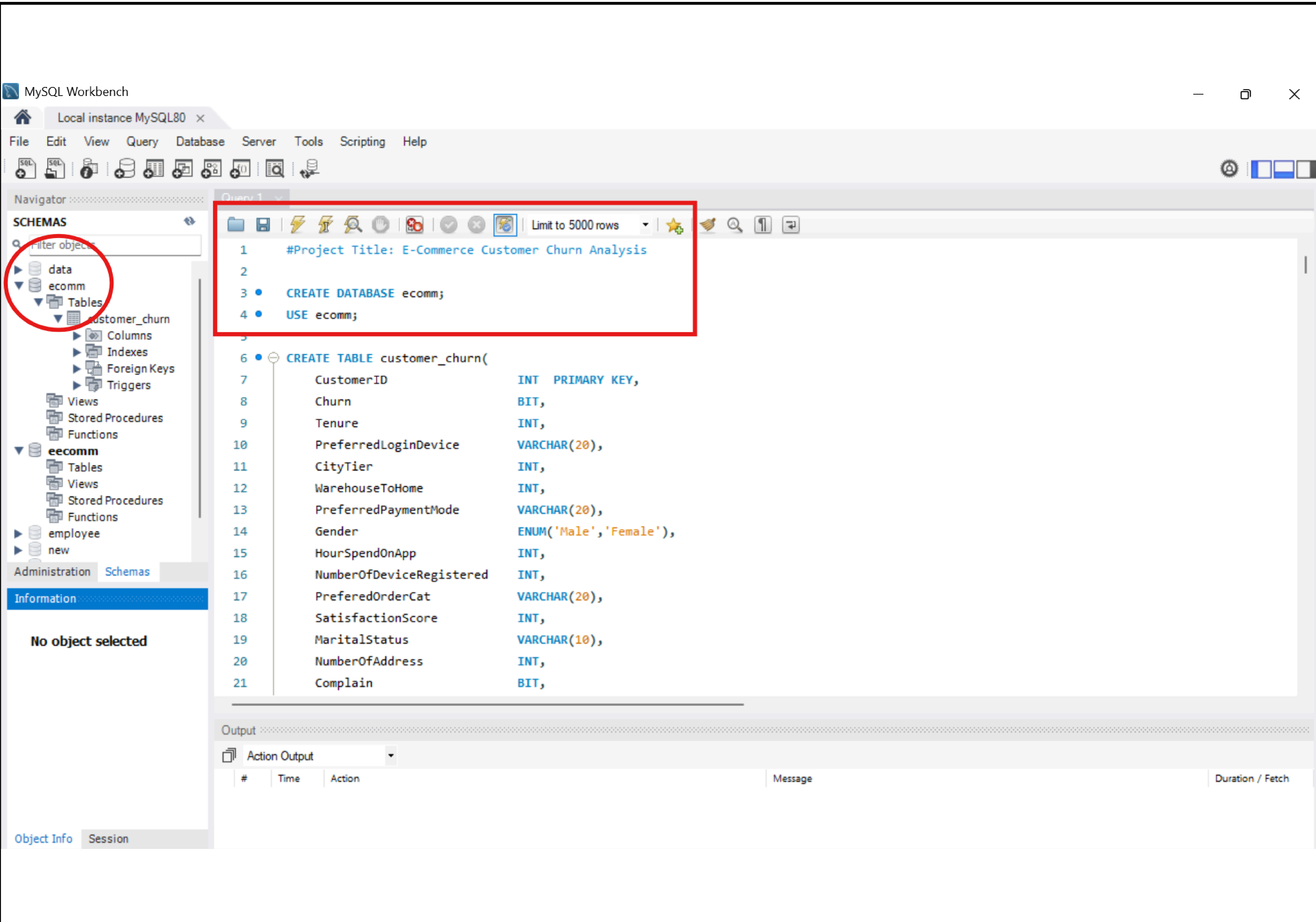
In the rapidly growing e-commerce industry, businesses face challenges in understanding customer churn patterns and identifying the underlying factors that lead customers to discontinue their purchases. This E-Commerce Customer Churn Analysis project uses MySQL to analyze customer data and identify important factors related to churn. By examining customer attributes such as tenure, preferred payment mode, order activity, satisfaction score, complaints, cashback, and warehouse-to-home distance, the project provides insights into customer behavior and helps businesses understand the factors contributing to churn. These insights can support effective strategies to improve customer retention and reduce churn.

Project Title: E-Commerce Customer Churn Analysis

Dataset Download:

https://drive.google.com/uc?export=download&id=1iKKCze_Fpk2n_g3BlZBiSjcDFdFcEn3D

Creating Dataset

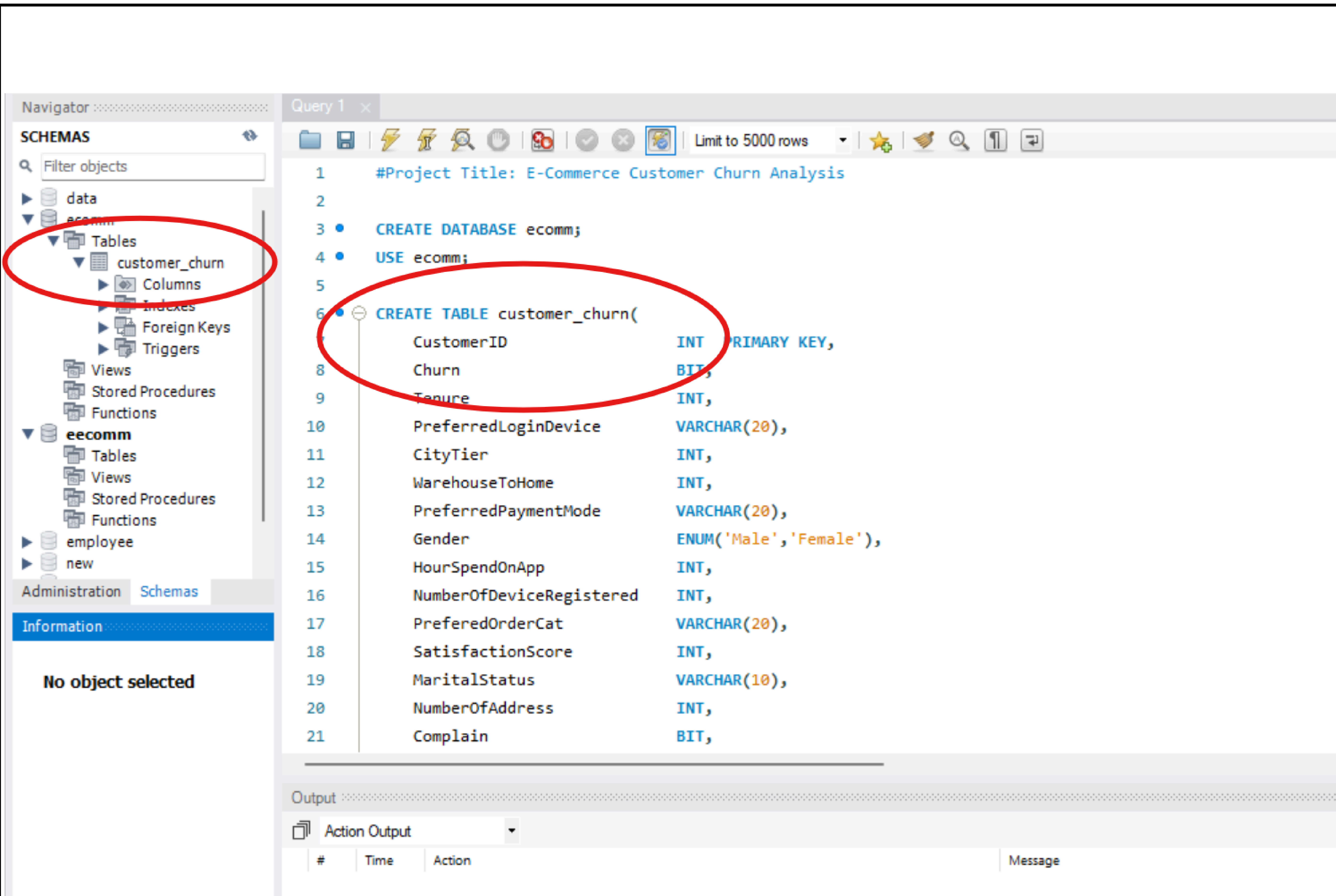


Project Title: E-Commerce Customer Churn Analysis

Dataset Download:

https://drive.google.com/uc?export=download&id=1iKKCze_Fpk2n_g3BlZBiSjcDFdFcEn3D

Creating Table customer_churn



Coulumns : CustomerID INT PRIMARY KEY,Churn BIT,Tenure INT,PreferredLoginDevice VARCHAR(20), CityTier INT, WarehouseToHome INT' PreferredPaymentMode VARCHAR(20) ,Gender ENUM('Male','Female'),HourSpendOnApp INT, NumberOfDeviceRegistered INT,PreferedOrderCat VARCHAR(20), SatisfactionScore INT,MaritalStatus VARCHAR(10),NumberOfAddress INT,Complain BIT,OrderAmountHikeFromlastYear INT,CouponUsed INT,OrderCount INT,DaySinceLastOrder INT,CashbackAmount INT.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

Tables

customer_churn

Columns

CustomerID

Churn

Tenure

PreferredLoginDe

CityTier

WarehouseToHon

PreferredPayment

Gender

HourSpendOnApp

NumberOfDeviceR

PreferedOrderCat

SatisfactionScore

MaritalStatus

Query 1

Limit to 5000 rows

```
5
6 CREATE TABLE customer_churn(
7     CustomerID          INT PRIMARY KEY,
8     Churn                BIT,
9     Tenure               INT,
10    PreferredLoginDevice VARCHAR(20),
11    CityTier              INT,
12    WarehouseToHome      INT,
13    PreferredPaymentMode  VARCHAR(20),
14    Gender                 ENUM('Male','Female'),
15    HourSpendOnApp        INT,
16    NumberOfDeviceRegistered INT,
17    PreferedOrderCat      VARCHAR(20),
18    SatisfactionScore     INT,
19    MaritalStatus         VARCHAR(10),
20    NumberOfAddress       INT,
21    Complain              BIT,
22    OrderAmountHikeFromlastYear INT,
23    CouponUsed            INT,
24    OrderCount            INT,
25    DaySinceLastOrder    INT,
26    CashbackAmount       INT
27 );
```

Output

Action Output

Time Action Message Duration / Fetch

Object Info Session

Data Cleaning: (5 marks)

Handling Missing Values and Outliers:

➤ Impute mean for the following columns, and round off to the nearest integer if required: WarehouseToHome, HourSpendOnApp, OrderAmountHikeFromlastYear, DaySinceLastOrder.

Mean Imputation for the "Warehouse to Home Column":

Navigator

SCHEMAS

Filter objects

customer_churn

Columns

CustomerID

Churn

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HourSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

Complain

OrderAmountHikeFromlastYear

Administration

Schemas

Information

Schema: ecomm

SQL File 1*

Limit to 1000 rows

5664 #Q-1 Handling Missing Values and Outliers:Impute mean for the following columns,WarehouseToHome, HourSpendOnApp, Order

5665 #DaySinceLastOrder.

5666

5667 • USE ecomm;

5668 • UPDATE customer_churn SET WarehouseToHome =

5669 (SELECT ROUND(AVG(Z.WarehouseToHome)) FROM (SELECT WarehouseToHome FROM customer_churn)AS Z)

5670 WHERE WarehouseToHome IS NULL;

5671 • SELECT * FROM customer_churn;

5672

5673

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

Fetch rows:

	CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrder
▶	50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Acces
	50002	1	NULL	Phone	1	8	UPI	Male	3	4	Mobile
	50003	1	NULL	Phone	1	30	Debit Card	Male	2	4	Mobile
	50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Acces
	50005	1	0	Phone	1	12	CC	Male	NULL	3	Mobile
	50006	1	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone
	50007	1	NULL	Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Acces

tomer_churn 1

Apply

Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✖ 9	15:29:19	UPDATE customer_churn SET WarehouseToHome = ROUND(AVG(WarehouseTo...	Error Code: 1111. Invalid use of group function	0.000 sec
✖ 10	15:42:58	UPDATE customer_churn SET WarehouseToHome = (SELECT ROUND(AVG(Z.W...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corr...	0.000 sec
✔ 11	15:44:58	UPDATE customer_churn SET WarehouseToHome = (SELECT ROUND(AVG(Z.W...	251 row(s) affected Rows matched: 251 Changed: 251 Warnings: 0	0.063 sec

Object Info

Session

Data Cleaning: (5 marks)

Handling Missing Values and Outliers:

➤ Impute mean for the following columns, and round off to the nearest integer if required: WarehouseToHome, HourSpendOnApp, OrderAmountHikeFromlastYear, DaySinceLastOrder.

Mean Imputation for the " OrderAmountHikeFromLastYear Column":

Navigator

SCHEMAS

Filter objects

CustomerID

Churn

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HourSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

Complain

OrderAmountHikeFromlastYear

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

Administration

Schemas

Information

Schema: ecomm

Object Info

Session

SQL File 1*

Limit to 1000 rows

WHERE HourSpendOnApp IS NULL;

#IMPUTE OrderAmountHikeFromlastyear

UPDATE Customer_churn SET OrderAmountHikeFromlastyear=

(SELECT ROUND(AVG(Z.OrderAmountHikeFromlastyear)) FROM (SELECT OrderAmountHikeFromlastyear FROM Customer_churn) AS Z)

WHERE OrderAmountHikeFromlastyear IS NULL;

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

Fetch rows:

	PreferedOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountHikeFromlastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
▶	laptop & Accessory	2	Single	9	1	11	1	1	5	160
	obile	3	Single	7	1	15	0	1	0	121
	obile	3	Single	6	1	14	0	1	3	120
	laptop & Accessory	5	Single	8	0	23	0	1	3	134
	obile	5	Single	3	0	11	1	1	3	130
	obile Phone	5	Single	2	1	22	4	6	7	139
	laptop & Accessory	2	Divorced	4	0	14	0	1	0	121

customer_churn 4

Apply

Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
17	16:31	UPDATE Customer_chum SET OrderAmountHikeFromlastyear= (SELECT ROUN...	265 row(s) affected Rows matched: 265 Changed: 265 Warnings: 0	0.032 sec
18	16:31:40	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec

Data Cleaning: (5 marks)

Handling Missing Values and Outliers:

➤ Impute mean for the following columns, and round off to the nearest integer if required: WarehouseToHome, HourSpendOnApp, OrderAmountHikeFromlastYear, DaySinceLastOrder.

Mean Imputation for the "HourSpendOnApp" Column:

Navigator

SCHEMAS

Filter objects

customer_churn

Columns

CustomerID

Churn

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HourSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

Complain

OrderAmountHikeFromlastYear

Administration

Schemas

Information

Schema: ecomm

Object Info

Session

SQL File 1*

Limit to 1000 rows

WHERE WarehouseToHome IS NULL;

SELECT * FROM customer_churn;

#IMPUTE HourSpendOnApp

UPDATE customer_churn SET HourSpendOnApp=

(SELECT ROUND(AVG(Z.HourSpendOnApp)) FROM (SELECT HourSpendOnApp FROM Customer_churn)AS Z)

WHERE HourSpendOnApp IS NULL;

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

Fetch rows:

	CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrder
▶	50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Acces
	50002	1	NULL	Phone	1	8	UPI	Male	3	4	Mobile
	50003	1	NULL	Phone	1	30	Debit Card	Male	2	4	Mobile
	50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Acces
	50005	1	0	Phone	1	12	CC	Male	3	3	Mobile
	50006	1	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone
	50007	1	NULL	Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Acces

customer_churn 3

Apply

Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 14	16:22:37	UPDATE customer_churn SET HourSpendOnApp= (SELECT ROUND(AVG(Z.Ho...	255 row(s) affected Rows matched: 255 Changed: 255 Warnings: 0	0.047 sec
✓ 15	16:24:48	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec

Data Cleaning: (5 marks)

Handling Missing Values and Outliers:

➤ Impute mean for the following columns, and round off to the nearest integer if required: WarehouseToHome, HourSpendOnApp, OrderAmountHikeFromlastYear, DaySinceLastOrder.

Mean Imputation for the "DaySinceLastOrder" Column:

SCHEMAS

Filter objects

CustomerID

Churn

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HourSpendOnApp

NumberOfDeviceRegistered

PreferedOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

Complain

OrderAmountHikeFromlastYear

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

Administration

Schemas

Information

Schema: ecomm

Object Info

Session

Limit to 1000 rows

5680

•

UPDATE Customer_churn SET OrderAmountHikeFromlastyear=

5681

(SELECT ROUND(AVG(Z.OrderAmountHikeFromlastyear)) FROM (SELECT OrderAmountHikeFromlastyear FROM Customer_churn) AS Z)

5682

WHERE OrderAmountHikeFromlastyear IS NULL;

5683

#IMPUTE DAYSINCE LAST ORDER

5684

•

UPDATE Customer_churn SET DaySinceLastOrder=

5685

(SELECT ROUND(AVG(Z.DaySinceLastOrder)) FROM (SELECT DaySinceLastOrder FROM Customer_churn)AS Z)

5686

WHERE DaySinceLastOrder IS NULL;

5687

5688

5689

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

Fetch rows:

	PreferedOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountHikeFromlastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
▶	laptop & Accessory	2	Single	9	1	11	1	1	5	160
	obile	3	Single	7	1	15	0	1	0	121
	obile	3	Single	6	1	14	0	1	3	120
	laptop & Accessory	5	Single	8	0	23	0	1	3	134
	obile	5	Single	3	0	11	1	1	3	130
	obile Phone	5	Single	2	1	22	4	6	7	139
	laptop & Accessory	2	Divorced	4	0	14	0	1	0	121

customer_churn 5

Apply

Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
19	16:38:32	UPDATE Customer_churn SET DaySinceLastOrder= (SELECT ROUND(AVG(Z.D...	307 row(s) affected Rows matched: 307 Changed: 307 Warnings: 0	0.062 sec
20	16:38:41	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.015 sec / 0.000 sec

Data Cleaning: (5 marks)

Handling Missing Values and Outliers:

➤ **Impute mode for the following columns: Tenure, CouponUsed, OrderCount.**

Mode Imputation for the "Tenur Column":

Navigator

SCHEMAS

Filter objects

CustomerID

Churn

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HourSpendOnApp

NumberOfDeviceRegistered

PreferedOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

Complain

OrderAmountHikeFromlastYear

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

Administration

Schemas

Information

Schema: ecomm

Object Info

Session

SQL File 1*

Limit to 1000 rows

5687 WHERE DaySinceLastOrder IS NULL;

5688

5689 #Impute mode for the following columns: Tenure, CouponUsed, OrderCount.

5690

5691 #IMPUTE Tenure WITH MODE

5692 UPDATE customer_churn SET Tenure=

5693 (SELECT Tenure FROM (SELECT Tenure FROM Customer_churn WHERE Tenure IS NOT NULL

5694 GROUP By Tenure ORDER BY COUNT(*) DESC LIMIT 1)AS Z)

5695 WHERE Tenure is NULL;

5696

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

Fetch rows:

	CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferedOrder
▶	50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Acces
	50002	1	1	Phone	1	8	UPI	Male	3	4	Mobile
	50003	1	1	Phone	1	30	Debit Card	Male	2	4	Mobile
	50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Acces
	50005	1	0	Phone	1	12	CC	Male	3	3	Mobile
	50006	1	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone
	50007	1	1	Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Acces

customer_churn 6

Apply

Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
23	17:02:11	UPDATE customer_churn SET Tenure= (SELECT Tenure FROM (SELECT Tenu...	264 row(s) affected Rows matched: 264 Changed: 264 Warnings: 0	0.016 sec
24	17:02:55	SELECT * FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec

Data Cleaning: (5 marks)

Handling Missing Values and Outliers:

➤ **Impute mode for the following columns: Tenure, CouponUsed, OrderCount.**

Mode Imputation for the "CouponUsed Column":

Navigator

SCHEMAS

Filter objects

- CustomerID
- Churn
- Tenure
- PreferredLoginDevice
- CityTier
- WarehouseToHome
- PreferredPaymentMode
- Gender
- HourSpendOnApp
- NumberOfDeviceRegistered
- PreferedOrderCat
- SatisfactionScore
- MaritalStatus
- NumberOfAddress
- Complain
- OrderAmountHikeFromlastYear
- CouponUsed
- OrderCount
- DaySinceLastOrder
- CashbackAmount

Administration Schemas

Information

Schema: ecomm

Object Info Session

SQL File 1*

Limit to 1000 rows

5694 GROUP By Tenure ORDER BY COUNT(*) DESC LIMIT 1)AS Z)

5695 WHERE Tenure is NULL;

5696

5697 #IMPUTE CouponUsed WITH MODE

5698 UPDATE Customer_churn SET CouponUsed=

5699 (SELECT CouponUsed FROM(SELECT CouponUsed FROM Customer_churn WHERE CouponUsed IS NOT NULL

5700 GROUP BY CouponUsed ORDER BY COUNT(*) DESC LIMIT 1)AS Z)

5701 WHERE CouponUsed IS NULL;

5702 SELECT*FROM customer_churn;

5703

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

Fetch rows:

	PreferedOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountHikeFromlastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
▶	laptop & Accessory	2	Single	9	1	11	1	1	5	160
	obile	3	Single	7	1	15	0	1	0	121
	obile	3	Single	6	1	14	0	1	3	120
	laptop & Accessory	5	Single	8	0	23	0	1	3	134
	obile	5	Single	3	0	11	1	1	3	130
	obile Phone	5	Single	2	1	22	4	6	7	139
	laptop & Accessory	2	Divorced	4	0	14	0	1	0	121

customer_churn 7

Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
25	17:07:59	UPDATE Customer_chum SET CouponUsed= (SELECT CouponUsed FROM(SE...	256 row(s) affected Rows matched: 256 Changed: 256 Warnings: 0	0.062 sec
26	17:08:37	SELECT*FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec



Data Cleaning: (5 marks)

Handling Missing Values and Outliers:

➤ **Impute mode for the following columns: Tenure, CouponUsed, OrderCount.**

Mode Imputation for the "OrderCount Column":

SCHEMAS

Filter objects

data

ecomm

Tables

customer_churn

customer_returns

Views

Stored Procedures

Functions

eecommm

employee

new

sys

testdb

Administration Schemas

Information

No object selected

Object Info Session

Limit to 5000 rows

#IMPUTE_OrderCount with MODE

UPDATE Customer_churn SET OrderCount=

(SELECT OrderCount FROM (SELECT OrderCount FROM Customer_churn WHERE

OrderCount IS NOT NULL

GROUP BY OrderCount ORDER BY COUNT(*) DESC LIMIT 1) AS Z)

WHERE OrderCount IS NULL;

COMMIT;

Result Grid

Filter Rows:

Edit

Export/Import

Fetch rows:

	SatisfactionScore	MaritalStatus	NumberOfAddress	OrderAmountHikeFromlastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount	ChurnStatus	ComplaintRecieved
2		Single	9	11	1	1	5	160	Churned	YES
3		Single	7	15	0	1	0	121	Churned	YES
3		Single	6	14	0	1	3	120	Churned	YES
5		Single	8	23	0	1	3	134	Churned	No
5		Single	3	11	1	1	3	130	Churned	No
5		Single	2	22	4	6	7	139	Churned	YES
2		Divorced	4	14	0	1	0	121	Churned	No
2		Divorced	3	16	2	2	0	123	Churned	YES
3		Divorced	2	14	0	1	2	127	Churned	YES
3		Single	2	12	1	1	1	123	Churned	No
3		Divorced	2	16	9	15	8	295	Churned	No

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	20:25:02	UPDATE Customer_churn SET OrderCount= (SELECT OrderCount FROM (SEL...	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0	0.031 sec
2	20:25:21	SELECT*FROM customer_churn LIMIT 0, 5000	5000 row(s) returned	0.000 sec / 0.000 sec

Data Cleaning: (5 marks)

Handling Missing Values and Outliers:

➤ **Handle outliers in the 'WarehouseToHome' column by deleting rows where the values are greater than 100.**

Navigator

SCHEMAS

Filter objects

data

ecomm

Tables

customer_churn

Columns

CustomerID

Churn

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HourSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCategory

SatisfactionScore

MaritalStatus

NumberOfAddress

Administration

Schemas

Information

Schema: ecomm

Object Info

Session

SQL File 1*

Limit to 5000 rows

OrderCount IS NOT NULL

GROUP BY OrderCount ORDER BY COUNT(*) DESC LIMIT 1) AS Z)

WHERE OrderCount IS NULL;

#Handle outliers in the 'WarehouseToHome' column by deleting rows where the values are greater than 100.

DELETE FROM Customer_churn

WHERE WarehouseToHome >100;

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

Fetch rows:

	CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCategory
▶	51307	0	5	Mobile Phone	3	6	E wallet	Female	2	3	Fashion
	51308	0	10	Mobile Phone	1	16	Debit Card	Female	2	1	Fashion
	51309	0	0	Phone	3	14	UPI	Female	2	3	Laptop & Accessories
	51311	0	17	Phone	1	16	CC	Male	2	3	Mobile
	51312	0	0	Phone	3	16	CC	Male	2	3	Mobile
	51313	0	12	Computer	1	15	Debit Card	Male	2	3	Others

customer_churn 3

Apply

Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	22:55:17	SELECT*FROM customer_churn WHERE WarehouseToHome >100 LIMIT 0, 1000	2 row(s) returned	0.016 sec / 0.000 sec
✓ 2	22:57:12	DELETE FROM Customer_churn WHERE WarehouseToHome >100	2 row(s) affected	0.047 sec
✓ 3	22:57:53	SELECT*FROM customer_churn LIMIT 0, 1000	1000 row(s) returned	0.016 sec / 0.000 sec

Query Completed

Dealing with Inconsistencies:

- Replace occurrences of “Phone” in the 'PreferredLoginDevice' column and “Mobile” in the 'PreferedOrderCat' column with “Mobile Phone” to ensure uniformity.

SCHEMAS

Filter objects

data

ecomm

Tables

customer_churn

Columns

CustomerID

Churn

Tenure

PreferredLo

CityTier

WarehouseT

PreferredPay

Gender

HourSpendC

NumberOfD

PreferredOrd

Satisfaction!

Query 1

Limit to 5000 rows

5710

5711

5712

5713

5714

5715

5716

5717

5718

5719

5720

#Replace occurrences of “Phone” in the 'PreferredLoginDevice' column and

#“Mobile” in the 'PreferedOrderCat' column with “Mobile Phone” to ensure uniformity.

UPDATE Customer_churn SET PreferredLoginDevice=

REPLACE(PreferredLoginDevice , 'Phone', 'Mobile Phone')

WHERE PreferredLoginDevice='Phone';

Result Grid

Filter Rows:

50001

50002

50003

50004

50005

50006

50007

CustomerID

Churn

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HourSpendOnApp

NumberOfDeviceRegistered

PreferredOrder

Mobile Phone

Mobile Phone

Mobile Phone

Mobile Phone

Mobile Phone

Computer

Mobile Phone

3

1

1

1

3

1

3

6

8

30

15

12

22

11

Debit Card

UPI

Debit Card

Debit Card

CC

Debit Card

Cash on Delivery

Female

Male

Male

Male

Male

Female

Male

3

3

2

2

3

5

2

3

4

4

3

5

3

Laptop & Acces

Mobile

Mobile

Laptop & Acces

Mobile

Mobile Phone

Laptop & Acces

customer_churn 2

Apply

Revert

Output

Action Output

#

Time

Action

Message

Duration / Fetch

20

00:23:29

UPDATE Customer_chum SET PreferredLoginDevice= REPLACE(PreferredLoginD...

1231 row(s) affected Rows matched: 1231 Changed: 1231 Warnings: 0

0.032 sec

21

00:23:39

SELECT*FROM customer_chum LIMIT 0, 5000

5000 row(s) returned

0.015 sec / 0.000 sec

Administration

Schemas

Information

Schema: eecomm

Object Info

Session

➤ **Standardize payment mode values: Replace "COD" with "Cash on Delivery" and "CC" with "Credit Card" in the PreferredPaymentMode column.**

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

data

ecomm

Tables

customer_churn

Columns

CustomerID

Churn

Tenure

PreferredLo

CityTier

WarehouseT

PreferredPay

Gender

HourSpendC

NumberOfD

PreferredOrd

Satisfaction!

Query 1

Limit to 5000 rows

5722

5723 #Standardize payment mode values: Replace "COD" with "Cash on Delivery" and

5724 #"CC" with "Credit Card" in the PreferredPaymentMode column.

5725 #USED Nested Subqueries

5726

5727 UPDATE Customer_churn SET PreferredPaymentMode=

5728 REPLACE(REPLACE(PreferredPaymentMode,'COD','Cash on Delivery'),

5729 'CC','Credit Card');

5730 SELECT * FROM Customer_churn;

5731

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

Fetch rows:

	redLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferedOrderCat	SatisfactionScore	MaritalStatus
	er	3	14	E wallet	Female	2	3	Fashion	2	Single
	er	1	6	Cash on Delivery	Male	2	3	Laptop & Accessory	3	Divorced
	Phone	1	13	Credit Card	Male	2	4	Laptop & Accessory	3	Divorced
	Phone	1	15	Credit Card	Female	3	4	Mobile Phone	4	Divorced
	Phone	3	10	E wallet	Male	3	4	Fashion	2	Divorced
	er	3	8	E wallet	Female	3	3	Mobile Phone	3	Divorced
	Phone	1	16	Credit Card	Male	3	3	Mobile Phone	2	Divorced

Customer_churn 4

Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 25	00:37:24	UPDATE Customer_chum SET PreferredPaymentMode= REPLACE(REPLACE(Pref...	638 row(s) affected Rows matched: 5628 Changed: 638 Warnings: 0	0.032 sec
✓ 26	00:37:54	SELECT * FROM Customer_chum LIMIT 0, 5000	5000 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

➤ **Rename the column "PreferedOrderCat" to "PreferredOrderCat".**

SCHEMAS

Filter objects

data

ecomm

Tables

customer_churn

Columns

CustomerID

Churn

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMethod

Gender

HourSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCategory

SatisfactionScore

Administration

Schemas

Information

No object selected

Object Info

Session

Limit to 5000 rows

5730 • SELECT * FROM Customer_churn;

5731 • COMMIT;

5732

5733 #Data Transformation:

5734 #Rename the column "PreferedOrderCat" to "PreferredOrderCat".

5735

5736 ALTER TABLE customer_churn RENAME COLUMN PreferedOrderCat TO PreferredOrderCat ;

5737

5738 #Rename the column "HourSpendOnApp" to "HoursSpentOnApp".

5739 #Create a new column named "ComplainBased" with value "Yes" if the complain value is the (Complain)

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

Fetch rows:

id	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountHikeFromlastYear	CouponUsed	OrderCount	Day
3		Laptop & Accessory	2	Single	9	1	11	1	1	5
4		Mobile Phone	3	Single	7	1	15	0	1	0
4		Mobile Phone	3	Single	6	1	14	0	1	3
4		Laptop & Accessory	5	Single	8	0	23	0	1	3
3		Mobile Phone	5	Single	3	0	11	1	1	3
5		Mobile Phone	5	Single	2	1	22	4	6	7
3		Laptop & Accessory	2	Divorced	4	0	14	0	1	0

customer_churn 1 x

Apply

Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	01:53:43	ALTER TABLE customer_churn RENAME COLUMN PreferedOrderCat TO Preferred...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.219 sec
✓ 2	01:53:50	SELECT * FROM Customer_churn LIMIT 0, 5000	5000 row(s) returned	0.000 sec / 0.015 sec

➤ Rename the column "HourSpendOnApp" to "HoursSpentOnApp".

SCHEMAS

Filter objects

data

ecomm

Tables

customer_churn

Columns

CustomerID

Churn

Tenure

PreferredLoginDevio

CityTier

WarehouseToHome

PreferredPaymentMc

Gender

HourSpendOnApp

NumberOfDeviceReg

PreferredOrderCat

SatisfactionScore

Administration

Schemas

Information

No object selected

Object Info

Session

Limit to 5000 rows

5735 • ALTER TABLE customer_churn RENAME COLUMN PreferedOrderCat TO PreferredOrderCat ;

5736

5737 #Rename the column "HourSpendOnApp" to "HoursSpentOnApp".

5738 • ALTER TABLE Customer_churn RENAME COLUMN HourSpendOnapp TO HoursSpendOnApp;

5739

5740

5741 #Create a new column named 'ComplaintReceived' with values "Yes" if the corresponding value in the 'Complain'

5742 # Create a new column named 'ChurnStatus'. Set its value to "Churned" if the corresponding value in the 'Chur

5743 #Drop the columns "Churn" and "Complain" from the table.

5744

Result Grid

Filter Rows:

Export/Import:

Wrap Cell Content:

Fetch rows:

	useToHome	PreferredPaymentMode	Gender	HoursSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain
		Debit Card	Female	3	3	Laptop & Accessory	2	Single	9	1
		UPI	Male	3	4	Mobile Phone	3	Single	7	1
		Debit Card	Male	2	4	Mobile Phone	3	Single	6	1
		Debit Card	Male	2	4	Laptop & Accessory	5	Single	8	0
		Credit Card	Male	3	3	Mobile Phone	5	Single	3	0
		Debit Card	Female	3	5	Mobile Phone	5	Single	2	1
		Cash on Delivery	Male	2	3	Laptop & Accessory	2	Divorced	4	0

Customer_churn 2

Apply

Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	02:00:22	ALTER TABLE Customer_chum RENAME COLUMN HourSpendOnapp TO Hours...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.032 sec
4	02:00:46	SELECT * FROM Customer_chum LIMIT 0, 5000	5000 row(s) returned	0.000 sec / 0.016 sec

Creating New Columns:

➤ Create a new column named 'ComplaintReceived' with values "Yes" if the corresponding value in the 'Complain' is 1, and "No" otherwise.

Navigator

SCHEMAS

Filter objects

HoursSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

Complain

OrderAmountHikeFromlastY

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

ComplaintRecieved

ChurnStatus

Indexes

Foreign Keys

Triggers

Administration

Schemas

Information

Schema: ecomm

Object Info

Session

SQLA3Churn*

Limit to 5000 rows

5743 • SELECT DISTINCT `Complain` FROM customer_churn;

5744

5745

5746 #Create a new column named 'ComplaintReceived' with values "Yes" if the corresponding value in the 'Complain' is 1, and "No" otherwise.

5747 • ALTER TABLE `customer\_churn` ADD COLUMN `ComplaintRecieved` VARCHAR(3);

5748 • UPDATE Customer_churn SET `ComplaintRecieved` =CASE WHEN `Complain`=1 THEN 'YES' ELSE 'No' END;

5749

5750

5751 # Create a new column named 'ChurnStatus'. Set its value to "Churned" if the corresponding value in the 'Churn' column is 1, and "Not Churned" otherwise.

5752 • ALTER TABLE `Customer\_churn` ADD COLUMN `ChurnStatus` VARCHAR(10);

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

Fetch rows:

	MaritalStatus	NumberOfAddress	Complain	OrderAmountHikeFromlastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount	ChurnStatus	ComplaintRecieved
▶	Single	9	1	11	1	1	5	160	Churned	YES
	Single	7	1	15	0	1	0	121	Churned	YES
	Single	6	1	14	0	1	3	120	Churned	YES
	Single	8	0	23	0	1	3	134	Churned	No
	Single	3	0	11	1	1	3	130	Churned	No
	Single	2	1	22	4	6	7	139	Churned	YES

customer_churn 16

Apply

Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 36	03:00:46	ALTER TABLE `customer_chum` ADD COLUMN `ComplaintRecieved` VARCHAR(3)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
✓ 37	03:00:53	UPDATE Customer_chum SET `ComplaintRecieved` =CASE WHEN `Complain`=1 T...	5628 row(s) affected Rows matched: 5628 Changed: 5628 Warnings: 0	0.203 sec
✓ 38	03:01:00	SELECT FROM customer_churn LIMIT 0, 5000	5000 row(s) returned	0.000 sec / 0.015 sec

Query Completed

➤ Create a new column named 'ChurnStatus'. Set its value to “Churned” if the corresponding value in the 'Churn' column is 1, else assign “Active”.

SCHEMAS

Filter objects

HoursSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

Complain

OrderAmountHikeFromlast

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

ComplaintReceived

ChurnStatus

Indexes

Foreign Keys

Triggers

Administration

Schemas

Information

Schema: ecomm

Object Info

Session

Limit to 5000 rows

Create a new column named 'ChurnStatus'. Set its value to “Churned” if the corresponding value in the 'Churn' column is 1, else assign “Active”.

ALTER TABLE `Customer\_churn` ADD COLUMN `ChurnStatus` VARCHAR(10);

UPDATE `customer\_churn` SET `ChurnStatus` = CASE WHEN `Churn` =1 THEN 'Churned' ELSE 'Active' END;

#Drop the columns "Churn" and "Complain" from the table.

ALTER TABLE `Customer\_churn`

DROP COLUMN `ChurnStatus`,

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

Fetch rows:

Apply

Revert

id	SatisfactionScore	MaritalStatus	NumberOfAddress	Complain	OrderAmountHikeFromlastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount	ChurnStatus
1	3	Divorced	4	1	11	1	2	2	146	Churned
2	2	Single	3	0	15	0	1	3	207	Churned
3	5	Single	8	0	22	1	1	4	142	Churned
4	4	Divorced	2	0	13	1	1	6	173	Active
5	3	Divorced	3	1	20	0	4	5	123	Active
6	4	Married	9	0	18	1	1	15	123	Active

customer_churn 15

Output

Action Output

#	Time	Action	Message	Duration / Fetch
33	02:52:59	ALTER TABLE `Customer_chum` ADD COLUMN `ChumStatus` VARCHAR(10)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.078 sec
34	02:53:06	UPDATE `customer_chum` SET `ChumStatus` = CASE WHEN `Chum` =1 THEN 'C...	5628 row(s) affected Rows matched: 5628 Changed: 5628 Warnings: 0	0.137 sec
35	02:53:13	SELECT * FROM customer_chum LIMIT 0, 5000	5000 row(s) returned	0.000 sec / 0.016 sec

Query Completed

Column Dropping:

➤ Drop the columns "Churn" and "Complain" from the table.

SCHEMAS

Filter objects

Columns

- CustomerID
- Tenure
- PreferredLoginDevice
- CityTier
- WarehouseToHome
- PreferredPaymentMode
- Gender
- HoursSpendOnApp
- NumberOfDeviceRegistered
- PreferredOrderCat
- SatisfactionScore
- MaritalStatus
- NumberOfAddress
- OrderAmountHikeFromlast
- CouponUsed
- OrderCount
- DaySinceLastOrder
- CashbackAmount

Administration Schemas

Information

Table:
customer_churn

Columns:
CustomerID
Tenure
PreferredLoginDevice
CityTier
WarehouseToHome

Object Info Session

Limit to 5000 rows

```
5753 • UPDATE `customer_churn` SET `ChurnStatus` = CASE WHEN `Churn` =1 THEN 'Churned' ELSE 'Active' END;
5754
5755
5756 #Drop the columns "Churn" and "Complain" from the table.
5757 • ALTER TABLE `Customer_churn`
5758 DROP COLUMN `Churn`,
5759 DROP COLUMN `Complain`;
5760
5761 • SELECT * FROM customer_churn;
5762
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

Fetch rows:

	CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat
▶	50001	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory
	50002	1	Mobile Phone	1	8	UPI	Male	3	4	Mobile Phone
	50003	1	Mobile Phone	1	30	Debit Card	Male	2	4	Mobile Phone
	50004	0	Mobile Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory
	50005	0	Mobile Phone	1	12	Credit Card	Male	3	3	Mobile Phone
	50006	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone

customer_churn 17

Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 38	03:01:00	SELECT * FROM customer_churn LIMIT 0, 5000	5000 row(s) returned	0.000 sec / 0.015 sec
✓ 39	03:08:46	ALTER TABLE `Customer_churn` DROP COLUMN `Churn`, DROP COLUMN `Comp...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.062 sec
✓ 40	03:08:50	SELECT * FROM customer_churn LIMIT 0, 5000	5000 row(s) returned	0.000 sec / 0.015 sec

Query Completed

Data Exploration and Analysis: (17 marks)

➤ Retrieve the count of churned and active customers from the dataset.

Navigator

SCHEMAS

Filter objects

Gender

HoursSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

OrderAmountHikeFromlast

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

ChurnStatus

ComplaintRecieved

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Administration

Schemas

Information

Table:
customer_churn

Columns:
CustomerID
Tenure
PreferredLoginDevice
CityTier
WarehouseToHome

Object Info

Session

SQLA3Churn*

Limit to 5000 rows

5762

5763 #Data Exploration and Analysis: (17 marks)

5764 #➤ Retrieve the count of churned and active customers from the dataset.

5765 #(By using conditional aggregation method)

5766

5767 •

SELECT SUM(`ChurnStatus` = 'churned') AS Churned_customers,

SUM(`ChurnStatus`='Active') AS Active_customers

FROM `Customer\_churn`;

5768

5769

5770

5771

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Churned_customers	Active_customers
▶	948	4680

Result 18 x

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
42	03:24:53	SELECT SUM('ChumStatus' = 'chumed') AS Chumed_customers, SUM('ChumedSt...	Error Code: 1054 - Unknown column 'ChumedStatus' in 'field list'	0.000 sec
43	03:25:28	SELECT SUM('ChumStatus' = 'chumed') AS Chumed_customers, SUM('ChumStatu...	1 row(s) returned	0.016 sec / 0.000 sec

Query Completed

➤ Display the average tenure and total cashback amount of customers who churned.

SCHEMAS

Filter objects

CustomerID

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HoursSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

OrderAmountHikeFromlast

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

ChurnStatus

Administration

Schemas

Information

Table:
customer_churn

Columns:
CustomerID
Tenure
PreferredLoginDevice
CityTier
WarehouseToHome

Object Info

Session

Limit to 5000 rows

5769 FROM `Customer\_churn`;

5770

5771 #> Display the average tenure and total cashback amount of customers who churned.

5772

5773 SELECT AVG(`Tenure`) AS Average_Tenure ,

5774 SUM(`CashbackAmount`) AS Total_cashbackAmount

5775 FROM `Customer\_churn` WHERE `ChurnStatus` = 'churned';

5776

5777 #> Determine the percentage of churned customers who complained.

5778 #> Identify the city tier with the highest number of churned customers whose preferred order category is Laptop & Accessor

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Average_Tenure	Total_cashbackAmount
▶	3.1762	152030

Result 19

Read Only

Output

Action Output

	time	Action	Message	Duration / Fetch
✓	43 03:25:28	SELECT SUM('ChumStatus' = 'chumed') AS Chumed_customers, SUM('ChumStatu...	1 row(s) returned	0.016 sec / 0.000 sec
✓	44 03:33:33	SELECT AVG('Tenure') AS Average_Tenure , SUM('CashbackAmount') AS Total_c...	1 row(s) returned	0.031 sec / 0.000 sec

➤ Determine the percentage of churned customers who complained.

SCHEMAS

Filter objects

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HoursSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

OrderAmountHikeFromlast

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

ChurnStatus

ComplaintRecieved

Indexes

Foreign Keys

Administration

Schemas

Information

Table:
customer_churn

Columns:
CustomerID
Tenure
PreferredLoginDevice
CityTier
WarehouseToHome

Object Info

Session

Limit to 5000 rows

5774 SUM(`CashbackAmount`) AS Total_cashbackAmount

5775 FROM `Customer\_churn` WHERE `ChurnStatus` = 'churned';

5776

5777 #➤ Determine the percentage of churned customers who complained.

5778

5779 SELECT ROUND(COUNT(CASE WHEN `ComplaintRecieved`='Yes' THEN 1 END)* 100.0/COUNT(*),2)

5780 AS Percentageof_churned_customerswhocomplained

5781 FROM Customer_churn WHERE `ChurnStatus` = 'churned';

5782

5783 #➤ Identify the city tier with the highest number of churned customers whose preferred order category is Laptop & Accessor

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Percentageof_churned_customerswhocomplained

53.59

Result 22

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
48	03:47:36	SELECT COUNT(CASE WHEN `ComplaintRecieved`='Yes' THEN 1 END)* 100.0/C...	1 row(s) returned	0.000 sec / 0.000 sec
49	03:49:09	SELECT ROUND(COUNT(CASE WHEN `ComplaintRecieved`='Yes' THEN 1 END)*...	1 row(s) returned	0.063 sec / 0.000 sec

Query Completed

➤ Identify the city tier with the highest number of churned customers whose preferred order category is Laptop & Accessory.

SCHEMAS

Filter objects

CustomerID

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HoursSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

OrderAmountHikeFromLast

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

ChurnStatus

Administration

Schemas

Information

Column: CityTier

Definition: CityTier int

Object Info

Session

Limit to 5000 rows

5782

5783 #➤ Identify the city tier with the highest number of churned customers whose preferred order

5784 #category is Laptop & Accessory.

5785

5786 • SELECT `CityTier`, COUNT(*) AS CityTierWith_Highest_numberof_ChurnedCustomers_whoPreferred_LaptopAndAccessory

5787 FROM `customer\_churn` WHERE `ChurnStatus` = 'churned'AND `PreferredOrderCat`='Laptop & Accessory'

5788 GROUP BY `CityTier` ORDER BY CityTierWith_Highest_numberof_ChurnedCustomers_whoPreferred_LaptopAndAccessory DESC

5789 LIMIT 1;

5790

5791

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	CityTier	CityTierWith_Highest_numberof_ChurnedCustome
▶	3	150

Result 24

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
❌ 51	04:12:39	SELECT `CityTier`, COUNT(*) AS CityTierWith_Highest_numberof_ChurnedCustome...	Error Code: 1054. Unknown column 'City Tier' in 'group statement'	0.000 sec
✅ 52	04:12:53	SELECT `CityTier`, COUNT(*) AS CityTierWith_Highest_numberof_ChurnedCustome...	1 row(s) returned	0.016 sec / 0.000 sec

➤ Identify the most preferred payment mode among active customers.

Navigator

SCHEMAS

Filter objects

customer_churn

Columns

CustomerID

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HoursSpendOnApp

NumberOfDeviceRegister

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

OrderAmountHikeFromla

CouponUsed

OrderCount

DaySinceLastOrder

Administration

Schemas

Information

Column: PreferredPaymentMode

Collation: utf8mb4_0900_ai_ci

Definition: PreferredPaymentMode

Object Info

Session

SQLA3Churn*

Limit to 5000 rows

5788 GROUP BY `CityTier` ORDER BY CityTierWith_Highest_numberof_ChurnedCustomers_whoPreferred_LaptopAndAccessory DESC

5789 LIMIT 1;

5790

5791 #Identify the most preferred payment mode among active customers.

5792

5793 • SELECT `preferredPaymentMode`,COUNT(*) AS Most_PREFERREDPaymentModeCount

5794 FROM `customer\_churn` WHERE `ChurnStatus` = 'Active'

5795 GROUP BY `preferredPaymentMode` ORDER BY Most_PREFERREDPaymentModeCount DESC LIMIT 1;

5796

5797 #Calculate the total order amount hike from last year for customers who are single and prefer mobile phones for ordering.

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

preferredPaymentMode

Most_PREFERREDPaymentModeCount

Debit Card

1956

Result 3

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	19:09:30	SELECT `preferredPaymentMode`,COUNT(*) AS Most_PREFERREDPaymentMode FRO...	1 row(s) returned	0.031 sec / 0.000 sec
3	19:10:29	SELECT `preferredPaymentMode`,COUNT(*) AS Most_PREFERREDPaymentModeCoun...	1 row(s) returned	0.015 sec / 0.000 sec

➤ Calculate the total order amount hike from last year for customers who are single and prefer mobile phones for ordering.

SCHEMAS

Filter objects

Columns

- CustomerID
- Tenure
- PreferredLoginDevice
- CityTier
- WarehouseToHome
- PreferredPaymentMode
- Gender
- HoursSpendOnApp
- NumberOfDeviceRegistered
- PreferredOrderCat
- SatisfactionScore
- MaritalStatus
- NumberOfAddress
- OrderAmountHikeFromLastYear
- CouponUsed
- OrderCount
- DaySinceLastOrder
- CashbackAmount

Administration

Schemas

Information

Column: PreferredPaymentMode

Collation: utf8mb4_0900_ai_ci

Definition: PreferredPaymentMode

Object Info

Session

Limit to 5000 rows

5793 • SELECT `preferredPaymentMode`,COUNT(*) AS Most_PREFERREDPaymentModeCount

5794 FROM `customer\_churn` WHERE `ChurnStatus` = 'Active'

5795 GROUP BY `preferredPaymentMode` ORDER BY Most PreferredPaymentModeCount DESC LIMIT 1;

5796

5797 #Calculate the total order amount hike from last year for customers who are single and prefer mobile phones for ordering.

5798 • SELECT SUM(`OrderAmountHikeFromLastYear`)AS TotalOrderAmount_WhoaresingleAndPreferMobile

5799 FROM `Customer\_churn` WHERE `MaritalStatus` = 'Single' AND `PreferredOrderCat` = 'Mobile Phone';

5800

5801

5802

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

TotalOrderAmount_WhoaresingleAndPreferMobile

12177

Result 4 x

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	19:10:29	SELECT `preferredPaymentMode`,COUNT(*) AS Most_PREFERREDPaymentModeCoun...	1 row(s) returned	0.015 sec / 0.000 sec
4	19:22:40	SELECT SUM(`OrderAmountHikeFromLastYear`)AS TotalOrderAmount_Whoaresingl...	1 row(s) returned	0.000 sec / 0.000 sec

➤ Find the average number of devices registered among customers who used UPI as their preferred payment mode.

Navigator

SCHEMAS

Filter objects

CustomerID

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HoursSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

OrderAmountHikeFromlastYea

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

ChurnStatus

Administration

Schemas

Information

Column:

PreferredPaymentMode

Collation:

utf8mb4_0900_ai_ci

Definition:

PreferredPaymentMode

Object Info

Session

SQLA3Churn*

Limit to 5000 rows

#Find the average number of devices registered among customers who used UPI as their preferred payment mode.

SELECT ROUND(AVG(`NumberOfDeviceRegistered`),2) AS Avg_ofdeviceRegisterd

FROM `customer\_churn`

WHERE `PreferredPaymentMode` = 'UPI';

#Determine the city tier with the highest number of customers.

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Avg_ofdeviceRegisterd
▶	3.72

Result 6

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	19:31:15	SELECT AVG('NumberOfDeviceRegistered') AS Avg_ofdeviceRegisterd FROM `cu...	1 row(s) returned	0.000 sec / 0.000 sec
6	19:32:38	SELECT ROUND(AVG('NumberOfDeviceRegistered'),2) AS Avg_ofdeviceRegisterd...	1 row(s) returned	0.031 sec / 0.000 sec

➤ Determine the city tier with the highest number of customers.

Navigator

SCHEMAS

Filter objects

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HoursSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

OrderAmountHikeFromlastYea

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

ChurnStatus

ComplaintRecieved

Administration

Schemas

Information

Column:

PreferredPaymentMode

Collation:

utf8mb4_0900_ai_ci

Definition:

PreferredPaymentMode

Object Info

Session

SQLA3Churn*

Limit to 5000 rows

5803

•

SELECT

ROUND

(

AVG

(

NumberOfDeviceRegistered

)

,

2

)

AS

Avg_ofdeviceRegisterd

FROM

customer_churn

WHERE

PreferredPaymentMode

=

'UPT'

;

5804

5805

5806

5807

5808

5809

5810

5811

#Determine the city tier with the highest number of customers.

SELECT CityTier ,Count(*) AS HighestNumberOfCustomers

FROM Customer_churn GROUP BY CityTier ORDER BY

HighestNumberOfCustomers DESC LIMIT 1;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	CityTier	HighestNumberOfCustomers
▶	1	3666

Result 8

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 8	19:44:07	SELECT 'CityTier' ,Count(*) AS CityWith_HighestNumberOfCustomers FROM 'Custo...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 9	19:44:31	SELECT 'CityTier' ,Count(*) AS HighestNumberOfCustomers FROM 'Customer_chur...	1 row(s) returned	0.015 sec / 0.000 sec

➤ Identify the gender that utilized the highest number of coupons.

Navigator

SCHEMAS

Filter objects

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HoursSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

OrderAmountHikeFromlastYear

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

ChurnStatus

ComplaintRecieved

Administration

Schemas

Information

Column:

PreferredPaymentMode

Collation:

utf8mb4_0900_ai_ci

Definition:

PreferredPaymentMode

Object Info

Session

SQLA3Churn*

Limit to 5000 rows

#Identify the gender that utilized the highest number of coupons.

SELECT `Gender`,SUM(CouponUsed) AS HighestNumberOfCouponUsed_Gender

FROM `customer\_churn` GROUP BY `Gender`

ORDER BY HighestNumberOfCouponUsed_Gender

DESC LIMIT 1;

#List the number of customers and the maximum hours spent on the app in each preferred order category.

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

Gender

HighestNumberOfCouponUsed_Gender

Male

5629

Result 9

Read Only

Output

Action Output

#

Time

Action

Message

Duration / Fetch

9

19:44:31

SELECT `CityTier`,Count(*) AS HighestNumberOfCustomers FROM `Customer_chur...

1 row(s) returned

0.015 sec / 0.000 sec

10

19:54:09

SELECT `Gender`,SUM(CouponUsed) AS HighestNumberOfCouponUsed_Gender F...

1 row(s) returned

0.016 sec / 0.000 sec

➤ List the number of customers and the maximum hours spent on the app in each preferred order category.

SCHEMAS

Filter objects

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HoursSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

OrderAmountHikeFromlast

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

ChurnStatus

ComplaintRecieved

Indexes

Administration Schemas

Information

Column: MaritalStatus

Collation: utf8mb4_0900_ai_ci

Definition: MaritalStatus varchar(10)

Object Info Session

SQLA3Churn*

Limit to 5000 rows

5817 DESC LIMIT 1;

5818

5819 #List the number of customers and the maximum hours spent on the app in each preferred order category.

5820

5821 SELECT `PreferredOrderCat` , COUNT(*) AS `NumberOfCustomers` , MAX(`HoursSpendOnApp`) AS MAX_HoursSpendOnTheApp

5822 FROM `Customer\_churn` GROUP BY `PreferredOrderCat`;

5823

5824

5825

5826

5827

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	PreferredOrderCat	NumberOfCustomers	MAX_HoursSpendOnTheApp
▶	Laptop & Accessory	2050	5
	Mobile Phone	2078	5
	Others	264	4
	Fashion	826	5
	Grocery	410	4

Result 1

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	14:48:53	SELECT 'PreferredOrderCat' , COUNT(*) AS 'NumberOfCustomers' , MAX('HoursSp...	Error Code: 1054. Unknown column 'PrefferedOrderCat' in 'group statement'	0.000 sec
4	14:49:09	SELECT 'PreferredOrderCat' , COUNT(*) AS 'NumberOfCustomers' , MAX('HoursSp...	5 row(s) returned	0.015 sec / 0.000 sec

➤ Calculate the total order count for customers who prefer using credit cards and have the maximum satisfaction score.

Navigator

SCHEMAS

Filter objects

Columns

CustomerID

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HoursSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

OrderAmountHikeFromlast

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

Administration

Schemas

Information

Column:
PreferredPaymentMode

Collation:
utf8mb4_0900_ai_ci

Definition:
PreferredPaymentMode

Object Info

Session

SQLA3Churn*

Limit to 5000 rows

5824

5825

5826

5827

5828

5829

5830

5831

5832

5833

#Calculate the total order count for customers who prefer using credit cards and

#have the maximum satisfaction score.

#{Used Inner Query)

SELECT SUM(`OrderCount`) AS `TotalOrderCount\_UsingCreditCard` , MAX(`SatisfactionScore`) AS `MAX\_SatisfactionScore`

FROM `Customer\_churn` WHERE `PreferredPaymentMode` = 'Credit Card' AND

`SatisfactionScore` = (SELECT MAX(`SatisfactionScore`) FROM Customer_churn);

SELECT * FROM customer_churn;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	TotalOrderCount_UsingCreditCard	MAX_SatisfactionScore
▶	1122	5

Result 6

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 18	15:14:23	SELECT SUM("OrderCount") AS "TotalOrderCount_UsingCreditCard" FROM "Custo...	1 row(s) returned	0.015 sec / 0.000 sec
✓ 19	15:19:37	SELECT SUM("OrderCount") AS "TotalOrderCount_UsingCreditCard" , MAX("Satisfa...	1 row(s) returned	0.000 sec / 0.000 sec

➤ What is the average satisfaction score of customers who have complained?

Navigator

SCHEMAS

Filter objects

NumberOfAddress

OrderAmountHikeFromlast

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

ChurnStatus

ComplaintRecieved

Indexes

Foreign Keys

Triggers

ns

red Procedures

ctions

n

ee

Administration

Schemas

Information

Column:
PreferredPaymentMode

Collation:
utf8mb4_0900_ai_ci

Definition:
PreferredPaymentMode

Object Info

Session

SQLA3Churn*

Limit to 5000 rows

5831`SatisfactionScore` = (SELECT MAX(`SatisfactionScore`) FROM Customer_churn);

5832

5833 • SELECT * FROM customer_churn;

5834

5835 #What is the average satisfaction score of customers who have complained?

5836

5837 • SELECT ROUND(AVG(`SatisfactionScore`),2) AS AVG_OfSatScore_WhoComplained

5838 FROM `Customer\_churn` WHERE `ComplaintRecieved` = 'Yes' ;

5839

5840 # List the preferred order category among customers who used more than 5 coupons.

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

AVG_OfSatScore_WhoComplained

3.00

Result 8

Output

Action Output

#	Time	Action	Message	Duration / Fetch
20	15:30:54	SELECT AVG(`SatisfactionScore`) AS AVG_OfSatScore_WhoComplained FROM `...	1 row(s) returned	0.016 sec / 0.000 sec
21	15:31:54	SELECT ROUND(AVG(`SatisfactionScore`),2) AS AVG_OfSatScore_WhoComplain...	1 row(s) returned	0.015 sec / 0.000 sec

➤ List the preferred order category among customers who used more than 5 coupons.

Navigator

SCHEMAS

Filter objects

data

ecommm

Tables

customer_churn

Columns

CustomerID

Tenure

PreferredLo

CityTier

WarehouseT

PreferredPay

Gender

HoursSpend

NumberOfD

PreferredOrc

Satisfaction!

MaritalStatu

NumberOfAc

OrderAmour

Administration

Schemas

Information

No object selected

SQLA3Churn*

Limit to 5000 rows

5835 #What is the average satisfaction score of customers who have complained?

5836

5837 • SELECT ROUND(AVG(`SatisfactionScore`),2) AS AVG_OfSatScore_WhoComplained

5838 FROM `Customer\_churn` WHERE `ComplaintRecieved`= 'Yes' ;

5839

5840 # List the preferred order category among customers who used more than 5 coupons.

5841

5842 • SELECT `CustomerID`,`PreferredOrderCat`, `CouponUsed` FROM `Customer\_churn` WHERE CouponUsed >5;

5843

5844 #List the top 3 preferred order categories with the highest average cashback amount.

5845

Result Grid

Filter Rows:

CustomerID PreferredOrderCat CouponUsed

50011 Others 9

50021 Laptop & Accessory 6

50022 Fashion 11

50086 Fashion 7

50094 Fashion 12

50127 Fashion 10

50137 Laptop & Accessory 6

50167 Laptop & Accessory 6

Output

Action Output

Time Action Message Duration / Fetch

1 13:32:14 SELECT `CustomerID`,`PreferredOrderCat`, `CouponUsed` FROM `Customer\_churn` ... 303 row(s) returned 0.000 sec / 0.000 sec

➤ List the top 3 preferred order categories with the highest average cashback amount.

Navigator

SCHEMAS

Filter objects

CustomerID

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HoursSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

OrderAmountHikeFromlast

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

ChurnStatus

Administration

Schemas

Information

Column:
PreferredPaymentMode

Collation:
utf8mb4_0900_ai_ci

Definition:
PreferredPaymentMode

Object Info

Session

SQLA3Churn*

Limit to 5000 rows

5841

5842 • `SELECT `PreferredOrderCat`, `CouponUsed` FROM `Customer_churn` WHERE CouponUsed >5;`

5843

5844 #List the top 3 preferred order categories with the highest average cashback amount.

5845

5846 • `SELECT `PreferredOrderCat`, ROUND(AVG (`CashBackAmount`),2) AS AVG_CashBack FROM `Customer_churn``

5847 `GROUP BY `PreferredOrderCat` ORDER BY AVG_CashBack DESC LIMIT 3;`

5848

5849 #Find the preferred payment modes of customers whose average tenure is 10 months and have placed more than 500 orders.

5850

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	PreferredOrderCat	AVG_CashBack
►	Others	304.45
	Grocery	266.24
	Fashion	210.40

Result 11

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 25	15:41:41	SELECT `PreferredOrderCat`, `CouponUsed` FROM `Customer_churn` WHERE Cou...	303 row(s) returned	0.015 sec / 0.000 sec
✓ 26	15:49:24	SELECT `PreferredOrderCat`, ROUND(AVG (`CashBackAmount`),2) AS AVG_CashB...	3 row(s) returned	0.031 sec / 0.000 sec

➤ Find the preferred payment modes of customers whose average tenure is 10 months and have placed more than 500 orders.

SCHEMAS

Filter objects

CustomerID

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HoursSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

OrderAmountHikeFromlast

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

ChurnStatus

Administration

Schemas

Information

Column: PreferredPaymentMode

Collation: utf8mb4_0900_ai_ci

Definition: PreferredPaymentMode

Object Info

Session

Limit to 5000 rows

5847 GROUP BY `PreferredOrderCat` ORDER BY AVG_CashBack DESC LIMIT 3;

5848

5849 #Find the preferred payment modes of customers whose average tenure is 10 months and have placed more than 500 orders.

5850

5851 • SELECT `PreferredPaymentMode` , ROUND(AVG(`Tenure`),2) AS AVG_Tenure,

5852 SUM(`OrderCount`) AS TotalOrders

5853 FROM `Customer\_churn` GROUP BY `PreferredPaymentMode`

5854 HAVING ROUND(AVG(`Tenure`),0)= 10

5855 AND SUM(`OrderCount`)>500;

5856

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	PreferredPaymentMode	AVG_Tenure	TotalOrders
▶	Debit Card	9.95	6802
	Credit Card	9.78	5409
	E wallet	10.20	1849

Result 18 x

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 33	16:18:13	SELECT `PreferredPaymentMode` , ROUND(AVG(`Tenure`),2) AS AVG_Tenure, SU...	5 row(s) returned	0.015 sec / 0.000 sec
✓ 34	16:24:11	SELECT `PreferredPaymentMode` , ROUND(AVG(`Tenure`),2) AS AVG_Tenure, SU...	3 row(s) returned	0.016 sec / 0.000 sec

➤ Categorize customers based on their distance from the warehouse to home such as 'Very Close Distance' for distances <=5km, 'Close Distance' for <=10km, 'Moderate Distance' for <=15km, and 'Far Distance' for >15km. Then, display the churn status breakdown for each distance category.

SCHEMAS

Filter objects

NumberOfAddress

OrderAmountHikeFromlast

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

ChurnStatus

ComplaintRecieved

Indexes

Foreign Keys

Triggers

ns

red Procedures

ctions

n

ee

Administration

Schemas

Information

Column: WarehouseToHome

Definition: WarehouseToHome int

Object Info

Session

Limit to 5000 rows

5857

5858

5859

5860

5861

5862

5863

5864

5865

5866

5867

5868

5869

#Categorize customers based on their distance from the warehouse to home such

#as 'Very Close Distance' for distances <=5km, 'Close Distance' for <=10km,

#'Moderate Distance' for <=15km, and 'Far Distance' for >15km. Then, display the churn status breakdown for each distance category.

SELECT CASE WHEN `WarehouseToHome` <=5 THEN 'Very Close Distance'

WHEN `WarehouseToHome` <=10 THEN 'Close Distance'

WHEN `WarehouseToHome` <=15 THEN 'Moderate Distance'

ELSE 'Far Distance'

END AS Distance_Cat, ChurnStatus, COUNT(*) AS Customer_Count

FROM `Customer\_churn`

GROUP BY Distance_Cat, `ChurnStatus`

ORDER BY Distance_Cat, `ChurnStatus`;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Distance_Cat	ChurnStatus	Customer_Count
▶	Close Distance	Active	1696
	Close Distance	Churned	265
	Far Distance	Active	1871
	Far Distance	Churned	498
	Moderate Distance	Active	1106
	Moderate Distance	Churned	184
	Very Close Distance	Active	7
	Very Close Distance	Churned	1

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
36	16:41:08	SELECT CASE WHEN 'WarehouseToHome' <=5 THEN 'Very Close Distance' WH...	Error Code: 1146. Table 'ecommm.custoer_chum' doesn't exist	0.000 sec
37	16:41:39	SELECT CASE WHEN 'WarehouseToHome' <=5 THEN 'Very Close Distance' WH...	8 row(s) returned	0.016 sec / 0.000 sec

➤ List the customer’s order details who are married, live in City Tier-1, and their order counts are more than the average number of orders placed by all customers.

Navigator

SCHEMAS

Filter objects

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HoursSpendOnApp

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

OrderAmountHikeFromlastY

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

ChurnStatus

ComplaintRecieved

Administration

Schemas

Information

Column:
OrderAmountHikeFromlastYe

Definition:
OrderAmountHikeFromlastYe

Object Info

Session

SQLA3Churn*

Limit to 5000 rows

#List the customer’s order details who are married, live in City Tier-1, and their
#order counts are more than the average number of orders placed by all customers.

SELECT `CustomerID` , `MaritalStatus` , `CityTier` , `OrderCount`
FROM `Customer\_churn`
WHERE `MaritalStatus` = 'Married' AND `CityTier` = 1
AND `OrderCount` >(SELECT ROUND(AVG(`OrderCount`),2)
FROM `Customer\_churn`);

Result Grid

Filter Rows:

	CustomerID	MaritalStatus	CityTier	OrderCount
▶	50049	Married	1	6
	50119	Married	1	4
	50132	Married	1	7
	50133	Married	1	3
	50152	Married	1	6
	50180	Married	1	3
	50187	Married	1	9
	50229	Married	1	6

Customer_churn 20

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✖ 41	17:03:56	SELECT `CutomerID` , `MaritalStatus` , `CityTier` , `OrderCount` FROM `Customer_ch...	Error Code: 1054. Unknown column 'CutomerID' in field list'	0.000 sec
✔ 42	17:04:08	SELECT `CustomerID` , `MaritalStatus` , `CityTier` , `OrderCount` FROM `Customer_c...	576 row(s) returned	0.016 sec / 0.000 sec

➤ a) Create a 'customer_returns' table in the 'ecomm' database and insert the following data:

ReturnID	CustomerID	ReturnDate	RefundAmount
1001	50022	2023-01-01	2130
1002	50316	2023-01-23	2000
1003	51099	2023-02-14	2290
1004	52321	2023-03-08	2510
1005	52928	2023-03-20	3000
1006	53749	2023-04-17	1740
1007	54206	2023-04-21	3250
1008	54838	2023-04-30	1990

Navigation

SCHEMAS

Filter objects

data

ecomm

Tables

customer_churn

customer_returns

Columns

ReturnID

CustomerID

ReturnDate

RefundAmount

Indexes

Foreign Keys

Triggers

Views

Stored Procedures

Functions

ecomm

employee

new

Administration

Schemas

Information

Schema: ecomm

Object Info

Session

SQLA3Churn*

Limit to 5000 rows

CREATE TABLE customer_returns (
ReturnID INT PRIMARY KEY,
CustomerID INT,
ReturnDate DATE,
RefundAmount INT,
FOREIGN KEY (CustomerID)
REFERENCES customer_churn(CustomerID));
INSERT INTO customer_returns (ReturnID, CustomerID, ReturnDate, RefundAmount)
VALUES
(1001, 50022, '2023-01-01', 2130),
(1002, 50316, '2023-01-23', 2000),
(1003, 51099, '2023-02-14', 2290),
(1004, 52321, '2023-03-08', 2510),
(1005, 52928, '2023-03-20', 3000),
(1006, 53749, '2023-04-17', 1740),
(1007, 54206, '2023-04-21', 3250),
(1008, 54838, '2023-04-30', 1990);

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	ReturnID	CustomerID	ReturnDate	RefundAmount
▶	1001	50022	2023-01-01	2130
	1002	50316	2023-01-23	2000
	1003	51099	2023-02-14	2290
	1004	52321	2023-03-08	2510
	1005	52928	2023-03-20	3000
	1006	53749	2023-04-17	1740
	1007	54206	2023-04-21	3250
	1008	54838	2023-04-30	1990

customer_returns 2

Apply

Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 3	14:16:02	CREATE TABLE customer_returns (ReturnID INT PRIMARY KEY, CustomerID INT, ReturnDate DATE, RefundAmount INT, FOREIGN KEY (CustomerID) REFERENCES customer_churn(CustomerID));	0 row(s) affected	0.188 sec
✓ 4	14:16:28	INSERT INTO customer_returns (ReturnID, CustomerID, ReturnDate, RefundAmount) VALUES (1001, 50022, '2023-01-01', 2130), (1002, 50316, '2023-01-23', 2000), (1003, 51099, '2023-02-14', 2290), (1004, 52321, '2023-03-08', 2510), (1005, 52928, '2023-03-20', 3000), (1006, 53749, '2023-04-17', 1740), (1007, 54206, '2023-04-21', 3250), (1008, 54838, '2023-04-30', 1990);	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0	0.016 sec
✓ 5	14:17:04	SELECT * FROM customer_returns LIMIT 0, 5000	8 row(s) returned	0.000 sec / 0.000 sec

used a SELECT query with INFORMATION_SCHEMA.KEY_COLUMN_USAGE to display the key information from both the customer_churn and customer_returns tables. The result shows that CustomerID in the customer_churn table is a Primary Key, and ReturnID in the customer_returns table is also a Primary Key. It also shows that CustomerID in the customer_returns table has the constraint name customer_returns_ibfk_1. The term IBFK refers to the automatically generated name of the Foreign Key constraint created by MySQL, which connects customer_returns.CustomerID to customer_churn.CustomerID.

Navigation

SCHEMAS

Filter objects

data

ecomm

Tables

customer_churn

customer_returns

Columns

ReturnID

CustomerID

ReturnDate

RefundAmount

Indexes

Foreign Keys

Triggers

Views

Stored Procedures

Functions

ecomm

employee

new

SQLA3Churn*

Limit to 5000 rows

5904

5905

5906

5907

5908

5909

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5911

5912

5913

5914

5915

5916

SELECT

TABLE_NAME,

COLUMN_NAME,

CONSTRAINT_NAME,

REFERENCED_TABLE_NAME,

REFERENCED_COLUMN_NAME

FROM INFORMATION_SCHEMA.KEY_COLUMN_USAGE

WHERE TABLE_SCHEMA = 'ecomm'

AND TABLE_NAME IN ('Customer_Churn', 'Customer>Returns')

AND COLUMN_NAME IN ('CustomerID', 'ReturnID');

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	TABLE_NAME	COLUMN_NAME	CONSTRAINT_NAME	REFERENCED_TABLE_NAME	REFERENCED_COLUMN_NAME
	customer_churn	CustomerID	PRIMARY		
	customer_returns	ReturnID	PRIMARY		
	customer_returns	CustomerID	customer_returns_ibfk_1	customer_churn	CustomerID

Schema: ecomm

KEY_COLUMN_USAGE 3

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	14:35:21	SELECT	Error Code: 1054. Unknown column 'CustomerID' in 'where clause'	0.016 sec
8	14:36:09	SELECT	3 row(s) returned	0.000 sec / 0.000 sec



b) Display the return details along with the customer details of those who have churned and have made complaints.

(An Index was created on the CustomerID Coulmn .Because indexing this column in the relevant tables can improve the performance of join operations, especially when working with larger datasets.)

Navigator

SCHEMAS

Filter objects

data

ecomm

Tables

customer_churn

customer_returns

Views

Stored Procedures

Functions

eecomm

employee

new

sys

testdb

SQLA3Churn*

Limit to 5000 rows

5941

5942

5943

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5945

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5951

5952

5953

5954

5955

CREATE INDEX idx_CustomerID

ON customer_returns(CustomerID);

SELECT Customer_returns.ReturnID,

Customer_returns.CustomerID,

Customer_churn.CustomerID,

Customer_churn.ChurnStatus,

Customer_churn.ComplaintRecieved

FROM Customer_returns

INNER JOIN Customer_churn

ON Customer_returns.CustomerID = Customer_churn.CustomerID

WHERE Customer_churn.ChurnStatus = 'Churned'

AND Customer_churn.ComplaintRecieved = 'Yes';

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	ReturnID	CustomerID	CustomerID	ChurnStatus	ComplaintRecieved
▶	1002	50316	50316	Churned	YES
	1004	52321	52321	Churned	YES
	1006	53749	53749	Churned	YES

Administration

Schemas

Information

No object selected

Result 2

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	22:05:09	CREATE INDEX idx_CustomerID ON customer_returns(CustomerID)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.141 sec
6	22:05:39	SELECT Customer_returns.ReturnID, Customer_returns.CustomerID, Cus...	3 row(s) returned	0.000 sec / 0.000 sec

Final Transformed Customer Churn Dataset

The final transformed dataset is displayed after completing the required data cleaning, transformation, and analysis steps. The table shows the updated customer data with the necessary modifications and newly created columns. This final output represents the completed and organized dataset after applying all the required SQL operations.

Table: customer_churn part 1

SQLA3Churn*

Limit to 5000 rows

5951

AND Customer_churn.ComplaintRecieved = 'Yes';

5952

5953

5954

select * FROM customer_churn;

5955

5956

	CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore	MaritalStatus	NumberC
	50007	1	Mobile Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Accessory	2	Divorced	4
	50008	1	Mobile Phone	1	6	Credit Card	Male	3	3	Mobile Phone	2	Divorced	3
	50009	13	Mobile Phone	3	9	E wallet	Male	3	4	Mobile Phone	3	Divorced	2
	50011	4	Mobile Phone	1	18	Cash on Delivery	Female	2	3	Others	3	Divorced	2
	50014	0	Mobile Phone	1	15	Credit Card	Male	3	4	Mobile Phone	3	Divorced	1
	50020	19	Mobile Phone	1	20	Debit Card	Female	3	3	Mobile Phone	4	Divorced	10
	50021	0	Mobile Phone	3	12	Debit Card	Male	3	5	Laptop & Accessory	3	Divorced	5
	50022	20	Mobile Phone	1	29	Credit Card	Female	3	3	Fashion	2	Divorced	2
	50024	0	Mobile Phone	3	26	Debit Card	Female	3	5	Laptop & Accessory	3	Divorced	4
	50027	8	Mobile Phone	3	6	E wallet	Male	3	3	Fashion	4	Divorced	2
	50028	1	Mobile Phone	3	12	E wallet	Male	2	3	Laptop & Accessory	3	Divorced	3
	50031	2	Computer	1	6	Cash on Delivery	Male	2	3	Laptop & Accessory	3	Divorced	2
	50032	0	Mobile Phone	1	13	Credit Card	Male	2	4	Laptop & Accessory	3	Divorced	3
	50033	30	Mobile Phone	1	15	Credit Card	Female	3	4	Mobile Phone	4	Divorced	5
	50034	13	Mobile Phone	3	10	E wallet	Male	3	4	Fashion	2	Divorced	8
	50035	1	Computer	3	8	E wallet	Female	3	3	Mobile Phone	3	Divorced	7
	50036	0	Mobile Phone	1	16	Credit Card	Male	3	3	Mobile Phone	2	Divorced	2
	50038	1	Mobile Phone	1	27	Credit Card	Male	3	3	Mobile Phone	2	Divorced	2
	50043	3	Mobile Phone	3	11	Debit Card	Female	2	3	Fashion	2	Divorced	3
	50046	0	Mobile Phone	1	23	Cash on Delivery	Male	3	3	Mobile Phone	4	Divorced	1
	50056	4	Mobile Phone	1	18	Credit Card	Female	3	3	Mobile Phone	2	Divorced	9

customer_churn 5

Apply

Revert

Output

The final transformed dataset is displayed after completing the required data cleaning, transformation, and analysis steps. The table shows the updated customer data with the necessary modifications and newly created columns. This final output represents the completed and organized dataset after applying all the required SQL operations.

Table: customer_churn part 2

SQLA3Churn*

Final Transformed Customer Churn Dataset

The final transformed dataset is displayed after completing the required data cleaning, transformation, and analysis steps. The table shows the updated customer data with the necessary modifications and newly created columns. This final output represents the completed and organized dataset after applying all the required SQL operations.

Table: customer_returns .

SQLA3Churn*

Limit to 5000 rows

5951

AND Customer_churn.ComplaintRecieved = 'Yes';

5952

5953

5954

Select * FROM customer_churn;

5955

SELECT * FROM customer_returns;

5956

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	ReturnID	CustomerID	ReturnDate	RefundAmount
▶	1001	50022	2023-01-01	2130
	1002	50316	2023-01-23	2000
	1003	51099	2023-02-14	2290
	1004	52321	2023-03-08	2510
	1005	52928	2023-03-20	3000
	1006	53749	2023-04-17	1740
	1007	54206	2023-04-21	3250
	1008	54838	2023-04-30	1990
*	NULL	NULL	NULL	NULL

Checking Null values

The Churn and Complaint columns were checked using "SELECT DISTINCT" to verify that they contain only the expected values 0 and 1 and no unexpected or "NULL" values. This validation is important before creating the corresponding status columns, as clean and consistent input values ensure that the status is generated correctly without errors or incorrect classifications.

Navigator

SCHEMAS

Filter objects

NumberOfDeviceRegistered

PreferredOrderCat

SatisfactionScore

MaritalStatus

NumberOfAddress

OrderAmountHikeFromlastYear

CouponUsed

OrderCount

DaySinceLastOrder

CashbackAmount

ChurnStatus

ComplaintRecieved

Indexes

Foreign Keys

Triggers

customer_returns

ews

ored Procedures

nctions

Administration Schemas

Information

Table: customer_churn

Columns:

CustomerID

Tenure

PreferredLoginDevice

CityTier

WarehouseToHome

PreferredPaymentMode

Gender

HoursSpendOnApp

Object Info Session

SQLA3Churn*

Limit to 5000 rows

6959

6960

SELECT * FROM customer_churn WHERE CustomerID IS NULL;

6961

SELECT * FROM customer_churn WHERE Tenure IS NULL;

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	23:50:27	SELECT * FROM customer_chum WHERE CustomerID IS NULL LIMIT 0, 5000	0 row(s) returned	0.000 sec / 0.000 sec
8	23:52:45	SELECT * FROM customer_chum WHERE Tenure IS NULL LIMIT 0, 5000	0 row(s) returned	0.016 sec / 0.000 sec
9	23:54:05	SELECT * FROM customer_chum WHERE PreferredLoginDevice IS NULL LIM...	0 row(s) returned	0.016 sec / 0.000 sec
10	23:54:12	SELECT * FROM customer_chum WHERE CityTier IS NULL LIMIT 0, 5000	0 row(s) returned	0.016 sec / 0.000 sec
11	23:54:18	SELECT * FROM customer_chum WHERE WarehouseToHome IS NULL LIMIT ...	0 row(s) returned	0.016 sec / 0.000 sec
12	23:54:23	SELECT * FROM customer_chum WHERE PreferredPaymentMode IS NULL LIM...	0 row(s) returned	0.015 sec / 0.000 sec
13	23:54:27	SELECT * FROM customer_chum WHERE Gender IS NULL LIMIT 0, 5000	0 row(s) returned	0.000 sec / 0.000 sec
14	23:57:59	SELECT * FROM customer_chum WHERE HoursSpendOnApp IS NULL LIMIT 0...	0 row(s) returned	0.000 sec / 0.000 sec
15	23:58:04	SELECT * FROM customer_chum WHERE NumberOfDeviceRegistered IS NUL...	0 row(s) returned	0.016 sec / 0.000 sec
16	23:58:08	SELECT * FROM customer_chum WHERE PreferredOrderCat IS NULL LIMIT 0, ...	0 row(s) returned	0.015 sec / 0.000 sec
17	23:58:12	SELECT * FROM customer_chum WHERE SatisfactionScore IS NULL LIMIT 0, ...	0 row(s) returned	0.000 sec / 0.000 sec
18	23:58:19	SELECT * FROM customer_chum WHERE MaritalStatus IS NULL LIMIT 0, 5000	0 row(s) returned	0.000 sec / 0.000 sec
19	23:58:23	SELECT * FROM customer_chum WHERE NumberOfAddress IS NULL LIMIT 0, ...	0 row(s) returned	0.015 sec / 0.000 sec
20	23:58:30	SELECT * FROM customer_chum WHERE OrderAmountHikeFromlastYear IS N...	0 row(s) returned	0.000 sec / 0.000 sec
21	23:58:35	SELECT * FROM customer_chum WHERE CouponUsed IS NULL LIMIT 0, 5000	0 row(s) returned	0.015 sec / 0.000 sec
22	23:58:39	SELECT * FROM customer_chum WHERE OrderCount IS NULL LIMIT 0, 5000	0 row(s) returned	0.016 sec / 0.000 sec
23	23:58:42	SELECT * FROM customer_chum WHERE DaySinceLastOrder IS NULL LIMIT 0...	0 row(s) returned	0.000 sec / 0.000 sec
24	23:58:46	SELECT * FROM customer_chum WHERE CashbackAmount IS NULL LIMIT 0, ...	0 row(s) returned	0.000 sec / 0.000 sec
25	23:58:49	SELECT * FROM customer_chum WHERE ChumStatus IS NULL LIMIT 0, 5000	0 row(s) returned	0.000 sec / 0.000 sec
26	23:58:53	SELECT * FROM customer_chum WHERE ComplaintRecieved IS NULL LIMIT 0...	0 row(s) returned	0.016 sec / 0.000 sec